Spectral Tails and Exact Concentration Logits

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Abstract

For the half-filled finite sine kernel, we express the concentration logit as a convergent sum over the spectrum of a singular differential operator after its first N eigenvalues have been removed. The finite Fourier subspace is proved to exhaust those first N modes. A Green-kernel trace identity then determines the logarithmic derivative of the deflated connection coefficient, with no undetermined entire factor. The resulting matrix identity is an absolutely convergent odd-power series with positive coefficients. It gives exact sign and monotonicity statements and explicit spectral-tail truncation bounds at fixed N. Formal center coefficients are compared with this representation. Their large-N interpretation is developed in the reciprocal-trace companions, and the fixed-interior quantum-period remainder is proved separately in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*.

1 Finite reference problem

Fix an integer $N \geq 1$. For $0 \leq j, k < N$, let

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad H_{j-1,j} = H_{j,j-1} = \frac{j(N-j)}{2}, \quad H_{jj} = 0. \tag{1}$$

This is the classical discrete concentration geometry used in the free-fermion entanglement problem [1]. We use

$$\mathcal{P}_N = \log B - \log(I - B), \qquad K_N = -\mathcal{P}_N, \tag{2}$$

with natural logarithms. The kernel is strictly between zero and one: its quadratic form and that of its complement are integrals of the squared absolute value of a nonzero finite Fourier polynomial over nonempty arcs. No physical clock or energy scale is assumed.

Put $a = \pi/2$, $I = (-a, a)$, $\alpha = (N^2 - 1)/4$, and

$$U_v(x) = \sum_{j=0}^{N-1} v_j e^{i[j-(N-1)/2]x}. \tag{3}$$

Direct Fourier summation gives

$$\mathcal{L}_N U_v = U_{Hv}, \qquad \mathcal{L}_N u = (\cos x\, u')' + \alpha \cos x\, u. \tag{4}$$

The outer Fourier coefficients vanish because $\alpha - m(m \pm 1) = 0$ at the appropriate endpoints of the frequency set. Integration by parts on I, where $\cos x$ vanishes at the endpoints, proves $HB = BH$. Since H has positive bonds, it has simple spectrum. Its real eigenvectors therefore give common eigenpairs $Hv = \beta v$, $Bv = \zeta v$, with $0 < \zeta < 1$.

The diagonal matrix $E_{jj} = (-1)^j$ obeys

$$EHE = -H, \qquad EBE = I - B. \tag{5}$$

Thus the spectrum of H is symmetric. Write $h_N = \|H\|$. The case $N = 1$ has $h_N = 0$ and $\mathcal{P}_N = 0$.

2 The differential spectrum and its first finite block

Let A_N be the self-adjoint Friedrichs operator associated with the closure on $C_c^\infty(I)$ of

$$\mathfrak{a}_N[f] = \int_I \cos x\, |f'|^2 \, \mathrm{d}x - \alpha \int_I \cos x\, |f|^2 \, \mathrm{d}x. \tag{6}$$

The differential expression is $A_N = -\mathcal{L}_N$. Its eigenfunctions use the regular, logarithm-free solution at both endpoints. We justify this choice below rather than selecting a different self-adjoint extension implicitly. The role of this condition in singular Sturm–Liouville problems is classical [2, 3].

Lemma 1 (Compact spectrum and elementary bounds)**.** *The spectrum of A_N is simple and discrete. In increasing order, with zero-based indices, write it as $\mu_0 < \mu_1 < \cdots$. For $c = 2/\pi^2$,*

$$c\, k(k+1) - \alpha \;\le\; \mu_k \;\le\; \frac{1}{2} k(k+1). \tag{7}$$

In particular its resolvent is trace class away from the spectrum.

Proof. For $|x| < a$,

$$\frac{\cos x}{a^2 - x^2} = \frac{1}{2} \frac{\sin[(a-x)/2]}{(a-x)/2} \frac{\sin[(a+x)/2]}{(a+x)/2}$$

lies between $2/\pi^2$ and $1/2$. Thus the positive derivative form has the same form domain and equivalent form norm as that of the rescaled Friedrichs Legendre operator $-[(a^2 - x^2)f']'$, whose eigenvalues are $k(k+1)$ [3]. That form domain embeds compactly into $L^2(I)$. The potential $-\alpha \cos x$ is bounded between $-\alpha$ and zero. The min–max principle now gives compactness and (7).

At either endpoint, the two local solution types of the eigen-equation are analytic and analytic times a logarithm plus analytic. The logarithmic type has infinite derivative energy $\int_0^\epsilon r\, |1/r|^2 \, \mathrm{d}r$ and is excluded by the form domain. The regular type belongs to the closure: a cutoff from zero to one over $\epsilon^2 < r < \epsilon$, linear in $\log r$, has derivative energy $O(1/|\log \epsilon|)$ and vanishing L^2 error. For these regular solutions the endpoint flux vanishes, so the eigen-equation also holds in the operator domain. The regular solution at the left endpoint is unique up to a constant; hence each eigenvalue is simple. This argument concerns eigenfunctions, not arbitrary form-domain functions, which need not be bounded at the endpoints. Finally (7) implies summability of the resolvent eigenvalues. □

Theorem 2 (Exact selection of the first N modes)**.** *The eigenvalues of A_N on its first N modes are exactly the negatives of the eigenvalues of H. Consequently*

$$\mu_{N-1} = h_N, \qquad \mu_N > h_N \ge 0. \tag{8}$$

Proof. Let V_N be the real-valued part of the centered Fourier space spanned by the functions in (4). Its real dimension is N; it is invariant under A_N, and its eigenfunctions are regular at both endpoints. Thus it supplies N distinct eigenvalues.

A real continuous function orthogonal to V_N must have at least N sign changes if it has only finitely many zeros. Indeed, if its sign-change points are $z_1, \ldots, z_r$ with $r < N$, then

$$q(x) = \prod_{\ell=1}^{r} \sin \frac{x - z_\ell}{2} \, \cos^{N-1-r} \frac{x}{2}$$

belongs to V_N. On I, it has exactly those sign changes. The product of the function and q has a fixed sign and nonzero integral, contradicting orthogonality.

The eigenfunction of index k has at most $k+1$ nodal domains. For completeness, its restrictions to nodal domains lie in the closed form domain and have disjoint supports, each with Rayleigh quotient μ_k. If there were more than $k+1$ such pieces, their span would contain a nonzero vector orthogonal to the first $k+1$ eigenfunctions. Its quotient would be both μ_k and at least μ_{k+1}, a

contradiction. Interior zeros are simple, and endpoint regularity prevents an accumulation there, so the sign-change argument applies.

An eigenfunction of index $k < N$ outside V_N would be orthogonal to the invariant finite space, and would have at least N sign changes, contradicting the nodal bound. Hence V_N contains the first N modes. Symmetry (5) identifies the largest selected value as h_N, and simplicity gives the strict inequality in (8). □

3 Connection coefficients and a resolvent trace

Let $u(x;b)$ solve

$$(\cos x\, u')' + (\alpha \cos x - b)u = 0, \qquad u(-a;b) = 1, \tag{9}$$

with the regular left endpoint condition. Let $v(x;b) = u(-x;b)$. At the right endpoint define $C_N(b)$ by

$$u(x;b) = C_N(b)v(x;b)\log(a-x) + \text{an analytic function}.$$

The regular right factor has value one at a. The normalized Frobenius recursion and continuation show that C_N is entire in b, and its constant Wronskian is

$$W = \cos x\,(uv' - u'v) = C_N(b). \tag{10}$$

In particular $C_N(b) = -2u(0;b)u'(0;b)$. Its zeros are precisely $b = -\mu_k$. They are simple: at a real zero, reflection gives $u(a) = h = \pm 1$, and parameter differentiation yields

$$\int_I u^2 \,\mathrm{d}x = -h\, C_N'(b) > 0. \tag{11}$$

There are no nonreal zeros, because a regular solution at both endpoints would give an eigenvalue of the self-adjoint operator.

Lemma 3 (Trace identity without a normalization ambiguity)**.** *For b outside the zeros of C_N,*

$$\frac{C_N'(b)}{C_N(b)} = \mathrm{Tr}(A_N + b)^{-1} = \sum_{k=0}^{\infty} \frac{1}{\mu_k + b}. \tag{12}$$

Proof. The Green kernel of $A_N + b$ is $G_b(x,y) = -u(\min(x,y);b)v(\max(x,y);b)/C_N(b)$. The minus sign makes the derivative jump of $-\cos x\, \partial_x G$ equal to one. Differentiating (9) with respect to b and pairing with v gives

$$\left[\cos x\,(u_b v' - u_b' v)\right]' = -uv.$$

The boundary values are zero at $-a$ and $C_N'(b)$ at a, so $C_N'(b) = -\int_I uv\,\mathrm{d}x$.

For real $b > \alpha$, the resolvent is positive and trace class. On closed interior subintervals its continuous positive kernel has trace equal to the integral of its diagonal. Exhausting I by those subintervals and using trace convergence for a positive trace-class operator gives $\mathrm{Tr}(A_N + b)^{-1} = \int_I G_b(x,x)\,\mathrm{d}x$. This proves the first equality there. Both sides extend meromorphically, so the equality holds throughout their common domain. The series equality is the spectral theorem with (7). □

Let $D_N(b) = \det(bI - H)$ and $R_N(b) = C_N(b)/D_N(b)$, with its removable values filled in. The first-N theorem and (12) give the exact cancellation

$$\frac{R_N'(b)}{R_N(b)} = \sum_{k=N}^{\infty} \frac{1}{\mu_k + b}. \tag{13}$$

In particular R_N is nonzero on $(-\mu_N, \mu_N)$. There is no unspecified exponential factor: (12) fixes the logarithmic derivative directly. Related Wronskian determinant methods are classical [4]; no regularized determinant is needed in this argument.

4 Exact positive expansion and truncation bounds

Theorem 4 (Spectral-tail formula)**.** *Define, for* $|b| < \mu_N$*,*

$$F_N(b) = 2\sum_{k=N}^{\infty} \operatorname{artanh}\frac{b}{\mu_k} = \sum_{k=N}^{\infty} \log\frac{\mu_k + b}{\mu_k - b}. \tag{14}$$

The series converges locally uniformly. At every common finite eigenmode,

$$\log\frac{\zeta}{1-\zeta} = F_N(\beta). \tag{15}$$

Consequently the following series converges absolutely in matrix operator norm:

$$\boxed{\mathcal{P}_N = 2\sum_{k=N}^{\infty} \operatorname{artanh}(H/\mu_k)} = 2\sum_{r=0}^{\infty} \frac{Z_N(2r+1)}{2r+1} H^{2r+1}, \tag{16}$$

$$Z_N(s) = \sum_{k=N}^{\infty} \mu_k^{-s}, \qquad s \geq 1.$$

Every coefficient on the right is strictly positive.

Proof. We first recover the finite concentration ratio from the normalized connections. At a selected root $b = \beta$, the Fourier solution divided by its nonzero left endpoint equals $u(x;\beta)$ and has right endpoint $h = \pm 1$. On the complementary arc, translation by π gives the same equation with parameter $-\beta$. The centered Fourier function satisfies $U(x+2\pi) = \sigma U(x)$, where $\sigma = (-1)^{N-1}$. Thus the complementary normalized endpoint value is σh. Parseval and (11) imply

$$\frac{\zeta}{1-\zeta} = \sigma\frac{C_N'(\beta)}{C_N'(-\beta)} = \frac{R_N(\beta)}{R_N(-\beta)}.$$

The last equality uses $D_N(-b) = (-1)^N D_N(b)$ and its derivative; it also covers the central zero for odd N by removable values.

Integrating (13) from $-b$ to b gives (14) as $\log|R_N(b)/R_N(-b)|$. The zero-free interval fixes the positive ratio, and $\sum_{k\geq N} 1/\mu_k < \infty$ justifies integration term by term. This proves (15). Because $h_N/\mu_N < 1$, the scalar artanh expansions converge uniformly on the whole finite spectrum. Their absolute norm sums are bounded by $2\sum_{k\geq N} \operatorname{artanh}(h_N/\mu_k) < \infty$. Functional calculus and absolute convergence prove (16). □

The function F_N is odd and strictly increasing. For $0 < b < \mu_N$, $F_N(b)/b$ is strictly increasing, by its positive Taylor coefficients. In particular $\operatorname{sgn}(2B - I) = \operatorname{sgn} H$, including the zero mode. For $N \geq 2$, (16) also gives zero entries between sites of equal parity and strictly positive entries between sites of opposite parity: every odd power of the positive-bond matrix H is entrywise nonnegative, and an odd-length path connects every opposite-parity pair. The signs reverse for K_N. These are finite-N conclusions. The leading scaling-limit power series in [1] is a related result with different coefficients.

Corollary 5 (Explicit omitted-spectrum bound)**.** *Let* $0 \leq b \leq h_N$ *and let* $M \geq N$ *be the first omitted index. If* $\rho = (\alpha + b)/[cM(M+1)] < 1$*, then*

$$\frac{4b}{M} \;\leq\; F_N(b) - 2\sum_{k=N}^{M-1} \operatorname{artanh}(b/\mu_k) \;\leq\; \frac{2b}{c(1-\rho)M}. \tag{17}$$

For negative b*, oddness gives the corresponding signed bounds. The upper bound with* $b = h_N$ *also bounds the matrix operator-norm truncation error in* (16)*.*

Proof. Use $z \leq \operatorname{artanh} z \leq z/(1-z)$ for $0 \leq z < 1$. The upper eigenvalue bound in (7) gives the lower estimate. For the upper estimate, $\mu_k - b \geq c(1-\rho)k(k+1)$ for $k \geq M$. Now $\sum_{k=M}^{\infty} 1/[k(k+1)] = 1/M$. The norm statement follows from the spectral theorem and monotonicity. □

The bound concerns the exact differential eigenvalues. Computed approximations require their own error enclosures; an ordinary Galerkin calculation does not automatically make this a numerical certificate.

Corollary 6 (Acceleration and a quantitative certificate)**.** *For* $0 \le b \le h_N$*, set* $g_b(\mu) = 2[\operatorname{artanh}(b/\mu) - b/\mu]$*. Then*

$$F_N(b) = 2bZ_N(1) + \sum_{k=N}^{\infty} g_b(\mu_k). \tag{18}$$

Let $M \ge N$*,* $d = c - \alpha/[M(M+1)] > 0$*, and* $q = b/[dM(M+1)] < 1$*. The omitted sum obeys*

$$0 \le \sum_{k=M}^{\infty} g_b(\mu_k) \le \frac{2b^3}{15(1-q^2)d^3M^5}. \tag{19}$$

If $|\widehat{Z} - Z_N(1)| \le \delta_Z$ *and* $|\hat{\mu}_k - \mu_k| \le \delta_k$ *for* $N \le k < M$*, with* $a_k = \hat{\mu}_k - \delta_k > b$*, the absolute error of the resulting finite approximation is at most*

$$2b\delta_Z + \sum_{k=N}^{M-1} \frac{2b^3\delta_k}{a_k^2(a_k^2 - b^2)} + \frac{2b^3}{15(1-q^2)d^3M^5}. \tag{20}$$

Proof. For $0 \le z \le q < 1$, the positive series gives $\operatorname{artanh} z - z \le z^3/[3(1-q^2)]$. For $k \ge M$, (7) gives $\mu_k \ge d\,k(k+1)$. Moreover

$$\frac{1}{[k(k+1)]^3} \le \frac{1}{5}\left(\frac{1}{k^5} - \frac{1}{(k+1)^5}\right),$$

since $(k+1)^5 - k^5 - 5k^2(k+1)^2 = 5k^2 + 5k + 1 > 0$. Summing proves (19). Finally $|g_b'(\mu)| = 2b^3/[\mu^2(\mu^2 - b^2)]$ decreases for $\mu > b$. The mean-value theorem and the triangle inequality give (20). □

5 Center law: exact structure and formal predictions

Let $J = 2H/N$, $Jv = \lambda v$, and $t = \lambda/N = 2\beta/N^2$. Write

$$\varepsilon = \log\frac{1-\zeta}{\zeta}, \qquad e = \varepsilon + N\eta(t), \qquad \eta(t) = \pi t\,{}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2).$$

The exact involution sends (t, ε, e) to $(-t, -\varepsilon, -e)$. Thus a leading relation involving $e/|\varepsilon|$ cannot have the same positive sign on both branches. Symmetry alone does not fix the sign or magnitude of e/ε.

The previously reduced formal coefficients [5] give

$$\begin{aligned} \eta(t) &= \pi t + \tfrac{\pi}{4}t^3 + O(t^5), & \varphi_1(t) &= \tfrac{\pi}{4}t + \tfrac{25\pi}{32}t^3 + O(t^5), \\ \varphi_3'(0) &= -\tfrac{17\pi}{64}, & \varphi_5'(0) &= \tfrac{169\pi}{256}. \end{aligned} \tag{21}$$

Therefore the proposed expansion $\varepsilon = -N\eta + \varphi_1/N + \varphi_3/N^3 + \cdots$ has the sign-consistent formal center law

$$e \sim \frac{\pi\lambda}{4N^2} \sim -\frac{\varepsilon}{4N^2}. \tag{22}$$

This coefficient calculation does not prove a plunge asymptotic: a fixed-interior expansion away from $t = 0$ cannot simply be evaluated along a sequence of modes tending to zero.

The tail representation gives an exact, distinguished interpolation with

$$F_N'(0) = 2Z_N(1). \tag{23}$$

Differentiating the *formal candidate* at zero consequently suggests the separate trace asymptotic

$$Z_N(1) \overset{\text{formal}}{\sim} \frac{\pi}{N} - \frac{\pi}{4N^3} + \frac{17\pi}{64N^5} - \frac{169\pi}{256N^7} + \cdots. \tag{24}$$

Neither the spectral period conjecture at discrete points nor a value-only remainder estimate alone justifies differentiating an interpolant. Equation (24) is an additional formal prediction to be tested and proved independently. For even N, it can be tested directly through $Z_N(1) = C_N'(0)/C_N(0)$, since D_N is even and nonzero at zero. For odd N, the removable-value formula is $Z_N(1) = C_N''(0)/(2C_N'(0))$.

One further exact formal coefficient. The linear center coefficient can be continued one order without assuming a spectral asymptotic. Set $p = \cos x$, $q = (p - 2t)/(4p)$, $r = 1/(4p^2)$, and solve the formal equation

$$W^2 + \tfrac{1}{2}(\log W)'' - \tfrac{1}{4}[(\log W)']^2 = N^2 q + r, \quad W = N\sqrt{q}\,(1 + a_1 N^{-2} + a_2 N^{-4} + a_3 N^{-6} + a_4 N^{-8} + \cdots). \tag{25}$$

Primes in (25) mean x derivatives. Coefficient comparison in the ring $\mathbb{Q}(p)[t]/(t^2)$, using $\partial_x^2 f = (1-p^2)f_{pp} - pf_p$, gives

$$\begin{aligned} a_4(p,0) &= \frac{4096p^6 - 143616p^4 + 493824p^2 - 375733}{128p^8}, \\ \partial_t a_4(p,0) &= -\frac{16p^8 - 50952p^6 + 825534p^4 - 2268425p^2 + 1553721}{16p^9}. \end{aligned} \tag{26}$$

The recursion fixes each new coefficient linearly, so these expressions are checked by substitution after the three preceding orders. For the counterclockwise period convention $\varphi_7 = i \oint \sqrt{q}\, a_4 \, \mathrm{d}x$ with $\mathrm{d}x = -\mathrm{d}p/\sqrt{1-p^2}$, a fixed circle $2|t| < R < 1$ yields

$$\varphi_7'(0) = \pi \operatorname*{Res}_{p=0} \frac{\partial_t a_4(p,0) - a_4(p,0)/p}{\sqrt{1-p^2}} = -\frac{50777\pi}{16384}. \tag{27}$$

This is an exact formal coefficient identity. The associated N^{-9} coefficient suggested for (24) is $50777\pi/16384 = 9.7363678083\ldots$; that spectral asymptotic is not proved by the residue calculation.

6 Scope and verification

Direct finite-matrix center-mode calculations at $N = 32, 64, 128, 256, 512, 1024$ give $-4N^2 e/\varepsilon$ increasing from 0.9993955762 to 0.9999992990 at 100-digit working precision. An independent connection calculation of $Z_N(1)$ at $N = 8, 16, 32, 64, 128$ gives the residual after the four displayed terms in (24), multiplied by N^9, as

$$8.7298271, \quad 9.4540545, \quad 9.6634554, \quad 9.7179853, \quad 9.7317624.$$

These values are consistent with the coefficient in (27). At $N = 128$, increasing precision from 130 to 180 digits and the Taylor degree from 600 to 800 changes the retained trace by less than 3.88×10^{-90}. These are numerical checks, not error enclosures or proofs of the limits.

The proofs above identify an exact finite reference function and give convergent spectral representations. They do not establish the large-N expansion of the remaining differential spectrum, the uniform quantum-period remainder, the full Wilson relative-entropy law, or a field-theory mass gap. The decomposition isolates a further precise target: estimate the positive traces $Z_N(2r+1)$ with uniform control sufficient to compare (16) to the declared period candidate. The reciprocal-trace, positive-moment and whole-line results supply the stated fixed-interior estimates. For the specified infinite-chain Wilson comparison at fixed positive scaled mass, Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* gives the finite part and an $O_\mu(1/\log L)$ remainder; Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* supplies the even-dimensional bound $\|E_N\| \le 7$. A moving-edge expansion remains open. Neither the finite-part theorem nor that norm bound promotes the formal coefficient calculation above to an additional spectral asymptotic. Separate reproduction material records numerical discretizations, precision comparisons, formal coefficient checks and their limitations.

Acknowledgments and provenance. This work develops the finite reference problem in *From Prephysical Hypotheses to Physical Models*. The earlier half-filled connection proposal was supplied in correspondence prepared with Claude (Anthropic). The present spectral-tail argument and checks were developed with ChatGPT (OpenAI). The wider program has also used Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer); no particular proof here is attributed to those systems. Assistance implies neither external peer review nor endorsement. No priority claim is made.

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